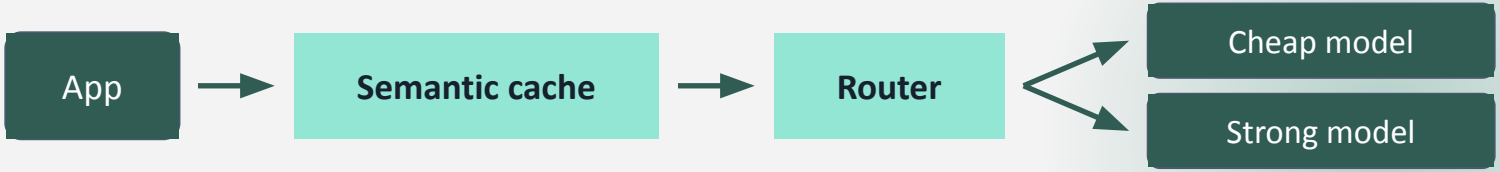
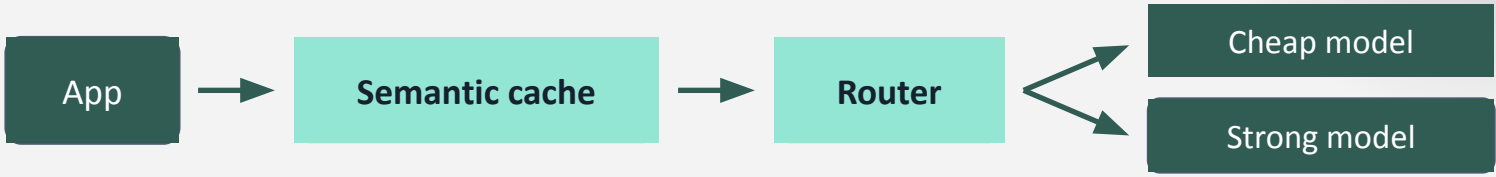
CacheShift
Sprint 2



Spencer Aldrich, Anthony Capraru
https://github.com/cacheshift-project/cacheshift

What CacheShift Does

- Many AI apps put a cache in front of a router: repeated questions get a saved answer, the rest go to a cheap or strong model
- We measure two things those tools don't report: **are the saved answers actually correct**, and **does the cache change how often the expensive model gets used**.
- Built for the person who sets up a small businesses AI feature and pays the bill
- Sprint 1 result: random repeats cause no shift. Mostly easy or mostly hard repeats cause drift



What Worked

Real-traffic Check

Scored 777k real chat prompts to see whether people's repeated questions differ from the one-off ones

Cached-answer quality

Built a semantic cache and graded every answer it reused against the official key

Blind-checked 50 and agreed with the grading on 48 (96%)

Re-tuning test

Re-ran the "fix the drift" command with a realistic, size-limited cache

GPU energy

Measured energy per token on shared GPU cluster; a 4-bit model used 43% less energy and ran 42% faster

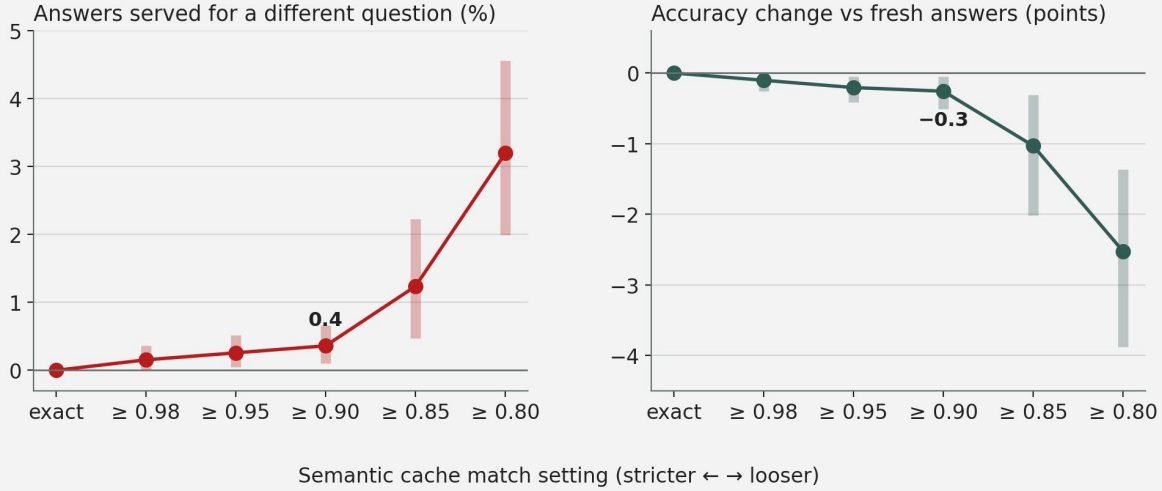
Target User

Narrowed to the builder of a small-budget AI feature

Demo

A cache that reuses answers to similar questions, showing when that goes right and wrong

Cached answers can be checked



gpt-4-1106-preview, seed 601 · 970 ARC questions + 970 repeats · answers graded against the official key · bars = 95% bootstrap ranges
blind hand check of 50 rows: 96% agreement · source: experiments/results/cache-quality-20261012

- The cache reuses an answer when a new question is “similar enough”; the setting decides how similar
- **Strict (≥ 0.90):** 0.4% of answers went to a different question, costing 0.3 points of accuracy
- **Loose (≥ 0.80):** 3.2% went to the wrong question, costing 2.5 points
- Errors stay near 0 until 0.90, then climb fast, so the setting matters
- Grading checked by hand: 48 of 50 agreed

Demo

```
CacheShift demo · a semantic cache in front of a bakery chatbot (made-up business)
6 saved answers · cache: sentence-transformers/all-MiniLM-L6-v2 · match settings: ≥ 0.80, ≥ 0.90, ≥ 0.95

1. "When do you open on Saturday?"
   closest saved question: "What time do you open on Saturdays?"   similarity 0.91
   ≥ 0.80 ✓ right answer   ≥ 0.90 ✓ right answer   ≥ 0.95 → to the model

   [Enter]
2. "Do you sell gluten free cookies?"
   closest saved question: "Do you have gluten-free cookies?"   similarity 0.92
   ≥ 0.80 ✓ right answer   ≥ 0.90 ✓ right answer   ≥ 0.95 → to the model

   [Enter]
3. "What time do you close on Sunday?"
   closest saved question: "What time do you close on Saturday?"   similarity 0.94
   ≥ 0.80 × WRONG answer   ≥ 0.90 × WRONG answer   ≥ 0.95 → to the model
   wrong answer served: "We close at 6 pm on Saturdays."

   [Enter]
4. "Are your muffins dairy-free?"
   closest saved question: "Are your muffins nut-free?"   similarity 0.83
   ≥ 0.80 × WRONG answer   ≥ 0.90 → to the model   ≥ 0.95 → to the model
   wrong answer served: "Yes, all our muffins are made without nuts."

   [Enter]
5. "Can I order a birthday cake for Friday?"
   closest saved question: "What time do you open on Saturdays?"   similarity 0.36
   ≥ 0.80 → to the model   ≥ 0.90 → to the model   ≥ 0.95 → to the model

   [Enter]
6. "Do you have gluten-free bread?"
   closest saved question: "Do you have gluten-free cookies?"   similarity 0.75
   ≥ 0.80 → to the model   ≥ 0.90 → to the model   ≥ 0.95 → to the model

Out of the 6 prepared questions:
   ≥ 0.80   2 right answers reused · 2 wrong answers served · 2 sent to the model
   ≥ 0.90   2 right answers reused · 1 wrong answer served · 3 sent to the model
   ≥ 0.95   0 right answers reused · 0 wrong answers served · 6 sent to the model
```

Energy is Measurable

6.28 J

Per token, full precision

3.59 J

Per token, 4-bit (43% less)

41.5 tok/s

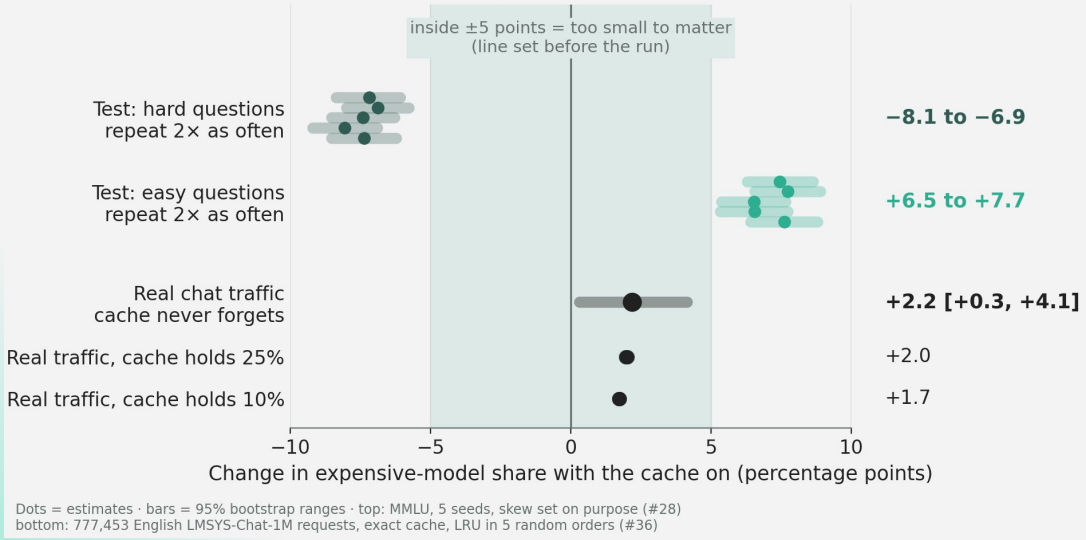
Full precision

59.0 tok/s

4-bit (42% faster)

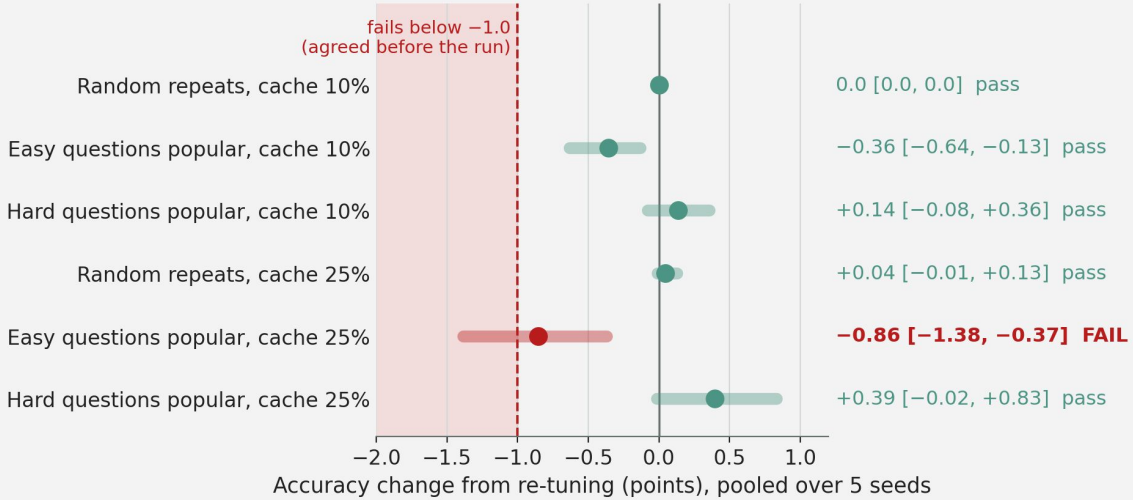
Llama-3.1-8B on one L40S GPU, one run on BU SCC

Real-traffic drift is small



- Expected: real repeats lopsided enough to move expensive-model use by 5+ points
- Got: +2.2 points on 777k real prompts (inside the line we set before running)
- People repeat slightly easier questions, so the cache takes the easy ones and the router sees a harder mix
- Half of the shift comes from 63 prompts asked 100+ times each
- **What Changed:** we measure how much it shifts on *your* traffic, not claim it always breaks

Re-tuning isn’t free



Dots = estimates · bars = paired 95% bootstrap ranges · every re-tuned share landed in 47–53%
MMLU, LRU exact cache · source: experiments/results/lru-retune-20261026

- Re-tuning brought expensive-model use back to plan (47-53%) in all 6 cache settings
- With easy questions popular and a large (25%) cache, accuracy dropped 0.86 points
- Why: the router now sees harder questions, so pulling it back to budget means fewer hard questions get to the strong model
- Open decision: keep the strict rule, or show users the trade-off

What we abandoned / didn’t do

The Headline

“Caching breaks routing budgets”; real traffic didn’t support it, so we now measure instead of warn

Plain random sampling

One prompt appeared 10,464 times and a random sample could miss it; we switched to sampling by repeat count before scoring anything

Cache in the live gateway

The semantic cache works in our offline tests but isn’t wired into the running gateway yet

Sprint 3

Demo Goal

The gateway produces the one-page report from a real run, with the semantic cache inside it

Decide

Whether re-tuning must never cost accuracy, or shows the trade-off for the user to pick

Questions?

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